

Lab 1 – Draft 1

Lydia Nebuchadnezzar

CS 411 – Thomas J. Kennedy

September 29, 2026

Version 2

Table of Contents

1 Introduction.....	2
2 Gold Star Product Description	2
2.1 Key Product Features and Capabilities	3
2.2 Major Components (Hardware/Software).....	3
3 Identification of Case Study.....	5
4 Glossary	6
5 References.....	8

Listing of Figures

Figure 1: Major Functional Component Diagram	4
--	---

1 Introduction

Individuals with ADHD diagnoses or ADHD-like traits often struggle with consistent routine development. ADHD is medically classified into three subtypes: Inattentive, Hyperactive-Impulsive, and Combined (Chhabildas et al., 2001, 529). With ADHD comes productivity-altering traits including inattention, hyperactivity, impulsivity, memory problems, and a lack of focus (Staley et al., 2024, p. 890; Advanced Psychiatry Associates, 2026).

ADHD often persists beyond childhood, as do the routine struggles that come with it. Routines may last anywhere from 3 days to “part of a month”, and are vulnerable to persistent Routine Adaptations (ADHD Philadelphia, 2026). Routine Adaptations often lead to frustration, anxiety, avoidance, and reduced self-efficacy (Chen et al., 2026, pp. 2, 7). Self-reliance on Routine Adaptations requires tedium creating new schedules while estimating uncertain timeframes. Individuals are also likely subject to error: they may forget to review their schedule, or Task Deviations may occur.

Gold Star is a web-based habit-building and time management platform tailored to those affected by ADHD. It is designed to help users organize their schedules, break down tasks, and build sustainable routines – making it easier to start, stay on track, and follow through every day.

2 Gold Star Product Description

Gold Star is a web-based habit-building platform designed for individuals struggling to maintain routines because of ADHD. Individuals diagnosed with ADHD commonly have difficulty creating routines, keeping habits, and staying on task. Gold Star offers a solution to these

problems by helping keep the user on track of what tasks need to get done every day with a flexible planner design.

2.1 Key Product Features and Capabilities

Gold Star is unique in that it utilizes an Adaptive Scheduling system to organize and schedule the user's tasks, eliminating the frustrating task deciding when to carry out daily obligations. In addition, Adaptive Scheduling considers varying energy and productivity levels, adjusting the number and difficulty of tasks for the day, while keeping important tasks that must be completed that day on the schedule. This flexibility is especially useful for people that struggle with ADHD-like symptoms.

Gold Star rewards users for completing tasks and milestones and does not punish users for missed or "Rolled Over" tasks. This approach gives users a sense of accomplishment and motivation to continue making progress. By supporting task initiation, consistency, commitment, and completion, Gold Star helps users stay organized, improve productivity, and reduce the stress associated with managing daily responsibilities.

2.2 Major Components (Hardware/Software)

The system does not require any specialized hardware. Users can access the application using any standard device with a modern web browser. The server side will operate in a single hosting environment that supports a Node.js web server and PostgreSQL database. Deployments will be automatically pushed through GitHub. As shown in Figure 1, the system is organized into a three-tier architecture consisting of a Presentation Layer, an Application Layer, and a Data Layer.

The Presentation Layer contains the web-based user interface. The Application Layer consists of the Node.js web server and API, which manage authentication, application logic, and notifications. The Data Layer uses a PostgreSQL database to store encrypted user information, tasks, and routine data.

The software being developed will include a user interface built with HTML, CSS, and JavaScript. The interface will provide features such as a calendar, routine dashboard, productivity tracking, and statistics dashboard. The Node.js web server will manage routing, application logic, authentication, and a WebSocket-based notification system. PostgreSQL will store user account information, tasks, routine data, and other sensitive information in an encrypted format. Development and testing will be supported by standard tools including Visual Studio Code, Git, GitHub, GitHub Actions for CI/CD, Jest for testing, and Javadoc for documentation.

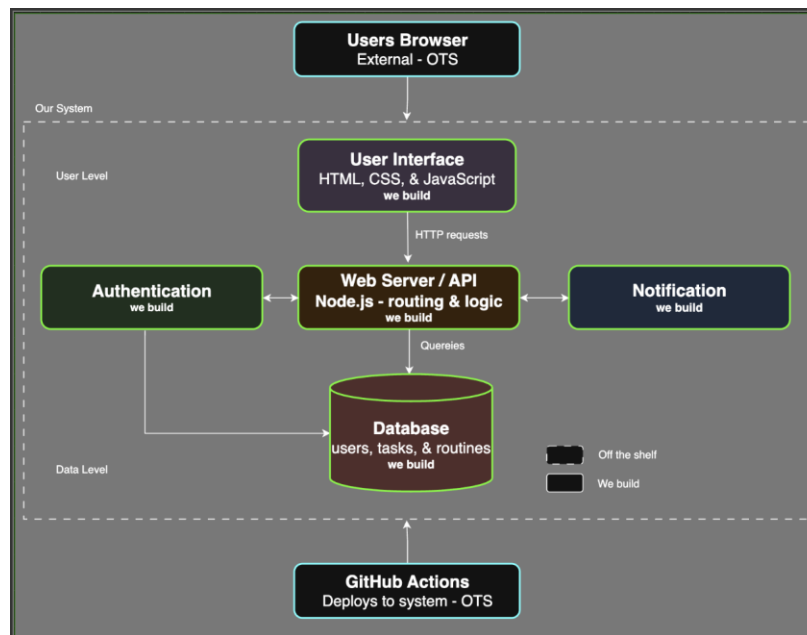


Figure 1: Major Functional Component Diagram

3 Identification of Case Study

For whom is this product being developed? Why? Who else might use this in the future?

[This space intentionally left blank]

4 Glossary

Activity Alternation – Switching from an activity requiring high mental focus to an activity requiring low mental focus.

Activity Switching – The cognitive process of disengaging from one task and shifting mental focus to another task.

Adaptive Scheduling – To schedule a task which can be completed anytime within a set timeframe. For example, a task which can be completed anywhere inclusively between 7:00 P.M. and 10:00 P.M.

Application Programming Interface (API) - A set of standard rules, protocols, and definitions allowing different software applications to interact with one another.

Attention-Deficit/Hyperactivity Disorder (ADHD) - A common neurodevelopmental disorder that alters the brain's ability to plan, focus, remember instructions, and manage emotions.

Cascading Style Sheets (CSS) - A style sheet language used to determine the visual presentation, layout, and aesthetics of a website.

Git - A digital version control system intended to preserve a history of saved modifications, additions, and deletions to software source code.

GitHub - A cloud-based platform where developers can collaborate on software code.

Gold Star - A web-based habit-building and time management platform designed specifically for individuals with ADHD.

Habit Stacking – Interspersing new habits with existing habits to make task completion easier without feeling overwhelmed.

Hyperactivity - A state of being unusually or abnormally active.

HyperText Markup Language (HTML) - The standard markup language used to create and structure webpages.

JavaScript (JS) - A lightweight programming language used to build dynamic interactivity on web pages.

Jest - An open-source JavaScript testing framework.

JSDoc - An API documentation generator and syntax standard for JavaScript.

Pomodoro Technique – A time-management technique which separates work into 25-minute increments and 5-minute breaks. The standard Pomodoro process repeats this technique 4 times, then a longer break between 20 to 30 minutes is taken before the process repeats.

PostgreSQL - An open-source RDBMS that stores, manages, and retrieves data.

Reinforce - To make something stronger, more effective, or more compelling by adding support.

Relational Database Management System (RBMS) - Software used to store, manage, query, and retrieve data structured in table form.

Rigid Scheduling – To schedule a high-priority task which must either be initiated at or completed before an established time. For example, a task which must start at 6:00 P.M. or must be completed by a deadline of 6:00 P.M.

Routine Adaption – A requirement for an individual to change one's schedule or create new routines.

Routine - A sequence of actions regularly followed, or a fixed program.

Task Deviation – When a task has an unexpected change regarding completion. A task may be accomplished out of order in one's schedule, or it may not be accomplished at all.

Visual Studio Code (VSCode) - A development environment used for software engineering and web development.

5 References

- ADHD Philadelphia. (2026, May 30). *Why Adults With ADHD Struggle to Stay Consistent With Routines*. ADHD Philadelphia. <https://www.adhdphiladelphia.com/blog/why-adults-with-adhd-struggle-to-stay-consistent-with-routines>
- Advanced Psychiatry Associates. (2026). *Strategies for Adults Living With ADHD*. Addicta. <https://advancedpsychiatryassociates.com/resources/blog/strategies-for-adults-living-with-adhd>
- CDC. (2024, May 29). *ADHD Across the Lifetime*. Attention-Deficit / Hyperactivity Disorder (ADHD). <https://www.cdc.gov/adhd/articles/adhd-across-the-lifetime.html>.
- Chen, J., Meng, Y., & Nie, K. (2026, March). “Not Just Me and My To-Do List”: Understanding Challenges of Task Management for Adults with ADHD and the Need for AI-Augmented Social Scaffolds. arXiv. <https://doi.org/10.48550/arXiv.2603.17258>
- Kessler, R.C., Lane, M., Stang, P.E., & Van Brunt, D.L. (2008, April 21). The prevalence and workplace costs of adult attention deficit hyperactivity disorder in a large manufacturing firm. *Psychological Medicine*, 39(1), 137-147. <https://doi.org/10.1017/S0033291708003309>
- Kreider, C. M., Medina, S., & Slamka, M. R. (2019). Strategies for Coping with Time-Related and Productivity Challenges of Young People with Learning Disabilities and Attention-Deficit/Hyperactivity Disorder. *Children (Basel, Switzerland)*, 6(2), 28. <https://doi.org/10.3390/children6020028>
- Ogrodnik, M., Karsan, S., Malamis, B., Kwan, M., Fenesi, B., & Heisz, J. J. (2023). Exploring Barriers and Facilitators to Physical Activity in Adults with ADHD: A Qualitative Investigation. *Journal of developmental and physical disabilities*, 1–21. Advance online publication. <https://doi.org/10.1007/s10882-023-09908-6>
- Staley, B.S., Robinson, L.R., Claussen, A.H., Katz, S.M., Danielson, M.L., Summers, A.D., Farr, S.L., Blumberg, S.J., & Tinker, S.C. (2024). Attention-Deficit/Hyperactivity Disorder Diagnosis,

Treatment, and Telehealth Use in Adults — National Center for Health Statistics Rapid Surveys System, United States, October–November 2023. *U.S. Centers for Disease Control and*

Prevention Morbidity and Mortality Weekly Report, 70(40), 890-895.

<https://www.cdc.gov/mmwr/volumes/73/wr/pdfs/mm7340a1-H.pdf>